VISVESVARAYA TECHNOLOGICAL UNIVERSITY

BELAGAVI

A Project Report on

LUNG CANCER DETECTION SYSTEM USING CNN

Submitted in partial fulfillment of the requirement for the award of the degree of

MASTER OF COMPUTER APPLICATIONS

Under

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

By

PAWAN KUMAR K.P

4SO23MC068

Department of Computer Applications

St Joseph Engineering College

Mangaluru-575028

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STJOSEPH ENGINEERING COLLEGE, MANGALURU

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CERTIFICATE

This is to certify that the project work titled

LUNG CANCER DETECTION SYSTEM USING CNN

SUBMITTED BY

PAWAN KUMAR K.P

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In partial fulfillment of the requirements for award of degree of Master of Computer Applications of Visvesvaraya Technological University, is a Bonafide record of the work carried out

During the academic year
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DECLARATION

I hereby declare that the entire work embodied in this dissertation has been carried out by me and no part of it has been submitted for any degree or diploma of any university previously.

Place: Mangaluru

Date: 31 July 2025

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Pawan Kumar

31 July 2025

ABSTARCT

The rapid advancement of Artificial Intelligence in medical diagnostics has significantly improved the early detection of life-threatening diseases. This project presents a web-based lung cancer detection system designed using a Convolutional Neural Network (CNN) integrated with the Flask web framework. The core objective is to assist healthcare professionals by automating the classification of lung CT scan images into three categories: benign, malignant, and normal.

The proposed system employs a hybrid 2×2 CNN model that extracts spatial features through convolutional and pooling layers, followed by a dense classifier to determine the final prediction. The model is trained on a labelled CT scan dataset with appropriate preprocessing steps such as resizing, normalization, and augmentation. The classification output is further visualized using a graphical bar chart to enhance interpretability for end users.

The application includes three main user roles: Admin, Doctor, and Patient. Admins manage users and system data, doctors can view and assess medical predictions, and patients can upload their CT scan images to receive immediate feedback. The entire system runs on a Flask backend with MySQL database integration, offering a user-friendly interface for image uploads, prediction results, and report generation.

This system reduces diagnostic delay by providing automated predictions and promotes accessibility by offering remote usage. Future enhancements may include real-time image capture integration, multilingual support, and expansion of the model to detect additional respiratory diseases. By combining deep learning with web technology, this project aims to provide a scalable, efficient, and accurate decision-support tool for early lung cancer screening.

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